Analysis2 - List 4: Questions on the Thoery of Fourier Series

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1 Orthogonal sequences and series

1. Definition of an abstract scalar product. Orthogonal vectors.
2. Definition of a norm of a vector induced by a scalar product.
3. Definition of a metric (distance) given by a norm.
4. Definition of a scalar product of functions square integrable at the given interval.
5. When do we say that two functions are orthogonal?
6. Does the orthogonality depend on the domain of the function? Provide an example.
7. Definition of the quadratic norm of the square integrable function.
8. Is the assumption about square integrability of a function important for a definition of a norm?
9. Definition of an orthogonal and orthonormal sequence of functions.
10. Examples of orthogonal and orthonormal sequence.
11. What is an orthogonal series?
12. What orthogonal sequence of functions is relevant for Fourier analysis?
13. Check that the given sequence of functions is orthogonal/orthonormal.
14. **Check that $1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \sin 3x \dots$ is orthogonal at $< -\pi, \pi >$.**
15. Check if the sequence of functions defined at the interval $< -l, l >$ is ortogonal/orthonormal: $1, \cos(\frac{\pi x}{l}), \sin(\frac{\pi x}{l}), \dots, \cos(\frac{n\pi x}{l}), \sin(\frac{n\pi x}{l}), \dots$
16. Explain the meaning of an orthogonal complete system of functions.
17. Provide an example of an orthogonal system of functions and some function, which does not belong to it.

2 Fourier series

18. **Formula for the Euler-Fourier coefficients at the interval $< -\pi, \pi >$ (ie.a_0, a_n, b_n).**
19. **Formula for the Euler-Fourier coefficients at the interval $< l, l >$, for some $l \in \mathbb{R}$.**

20. Definition of a Fourier series (in a general setting of some orthogonal complete system of functions (ϕ_n) defined at the interval $< -l, l >$).
21. Formula for the Fourier coefficients in a general setting of some orthogonal complete system of functions (ϕ_n).
22. The Euler-Fourier theorem for a general orthogonal complete system of functions.
23. **Rewrite The Euler-Fourier theorem for the orthogonal complete system of harmonics $1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \sin 3x \ldots$.**
24. **Compare the general formula the Fourier coefficients in case of (ϕ_n) with the specific formulae for a_n and b_n. Hint: How do we define scalar product of two square integrable functions?**
25. **Definition of a Fourier series (for a specific orthogonal complete system of harmonics ie. rescaled sinuses and cosines).**
26. **Which functions can be represented by a Fourier series? What do you mean by 'represented'?**
27. **Can a nonperiodic function be represented by a Fourier series?**
28. **List Dirichlet's conditions.**
29. **Under what conditions can we obtain pointwise convergence of a Fourier series? List those conditions.**
30. **Under what conditions can we obtain mean square convergence of a Fourier series?**
31. Under what conditions can we obtain uniform convergence of a Fourier series?
32. **Let f be a function which satisfies Dirichlet conditions, and let x_0 be its discontinuity point. What can you say about the convergence of the Fourier series at the point x_0?**
33. Discuss the relation between a Fourier series of a given integrable function f and the underlying function f.
34. **Explain the meaning of the Parseval condition. What does it tell about a function and its Fourier series?**
35. **Rewrite Parseval condition for the sequence of rescaled cosines and sinses. Hint: How do we define scalar product of two square integrable functions?**
36. **Let f be an odd function defined on $< -l, l >$. What can you say about its Fourier coefficients?**
37. **Let f be an even function defined on $< -l, l >$. What can you say about its Fourier coefficients?**
38. Is there any relation between above two questions and the last four questions form the section ”‘Orthogonal sequences and series”’?

39. **Calculating the Fourier coefficients for a given function (simple examples).**

3 The Least Square Approximation

40. **State the problem of the least square approximation.**

41. What norm is used in the Theorem of the Least Square Approximation? Provide the definition of this norm and the metric induced by this norm.

42. What is an orthogonal polynomial of order N?

43. How do we measure the distance between an orthogonal polynomial and the function related to it?

44. **Explain the meaning of the Bessel inequality.**

45. How do we measure the distance between an orthogonal polynomial and the Fourier series related to it?

46. **How is the least square approximation related to the Fourier series of the given function?**

47. **Theorem about the least square approximation.** (with the proof)

48. Quote the formulae for the coefficients in the least square approximation of a square integrable, periodic function.